

# Hoang Minh Thanh

## Machine Learning Engineer

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I focus on full-stack development of LLMs, from deep learning research knowledge to fine-tuning and production deployment. Adapting state-of-the-art open-source models using parameter-efficient methods (LoRA, QLoRA). I bring a combination of fundamental research depth and engineering execution.

## ★ Keypoint

- **Solid fundermental knowledge of Deep Learning:** Proven by my [thesis report](#), creator of [SurveyWithCode](#), proposal [DeepGesture](#). Experience in building evaluation pipelines for [perplexity](#), toxicity, and relevance to ensure safe and high-quality model outputs.
- **Experience in LLM Fine-Tuning & Distill & LoRA, QLoRA:** Proven by fine-tune F5-TTS model to create [Par-rotSpeech](#) TTS in Vietnamese, develop Chat LLM [MillionScope](#). Experience in SGLang, export ONNX, TensorRT, deploying models in production to prevent drift and ensure high-performance, reliable service.
- **Skilled in Prompt Engineering & Data Curation:** Adept at engineering prompts and curating datasets that reflect precise tone, brand voice, and safety guidelines, ensuring model outputs are both trustworthy and aligned with product objectives.
- **Problem solving and modern AI optimization technique:** FlashAttention, Attention Sink, KV Cache, CUDA Kernel C++

## 🎓 Education

|     |                                                                                                                                                                                                                                 |                         |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| AS  | University of Science - VNUHCM, Information Technology                                                                                                                                                                          | 2012 – 2015             |
| BS  | University of Science - VNUHCM, Computer Science                                                                                                                                                                                | Sept. 2018 – Sept. 2020 |
|     | <ul style="list-style-type: none"><li>• GPA: 3.3/4.0</li><li>• <b>Thesis:</b> GCAT - Link Prediction in Knowledge Graph, <a href="#">[Report]</a>, <a href="#">[Code]</a></li></ul>                                             |                         |
| MSc | University of Science - VNUHCM, Computer Science                                                                                                                                                                                | Oct. 2021 – Dec. 2024   |
|     | <ul style="list-style-type: none"><li>• GPA: 3.44/4.0</li><li>• <b>Thesis:</b> OpenHuman: A conversational gesture synthesis system based on emotions and semantics, <a href="#">[Report]</a>, <a href="#">[Code]</a></li></ul> |                         |

## 📖 Publications

|                                                                                                                                             |           |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| DeepGesture: A conversational gesture synthesis system based on emotions and semantics <a href="#">[arXiv]</a> , <a href="#">[homepage]</a> | Jul. 2025 |
| Towards a GENE Leaderboard <a href="#">[arXiv]</a>                                                                                          | Oct. 2024 |
| Graph Collaborative Attention Network for Link Prediction in Knowledge Graphs <a href="#">[arXiv]</a> <a href="#">[homepage]</a>            | Oct. 2020 |

## 💼 Experience

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Rakumo Inc, Software Engineer                                                                                                                                                                            | Jul. 2019 – Apr. 2020 |
| <ul style="list-style-type: none"><li>• Developed a Django-based web application with integrated DocuSign.</li></ul>                                                                                     |                       |
| FPT Software, AWS Data Engineer                                                                                                                                                                          | Oct. 2020 – Nov. 2021 |
| <ul style="list-style-type: none"><li>• CI/CD pipeline terraform in Jenkins, implemented and modified real-time and batch data pipelines on AWS using Kinesis, Lambda, S3, Glue, StepFunction,</li></ul> |                       |

SNS, ..etc..

- Cox Automotive: Collaborated with international teams in Agile

**VNG Corp**, Software Engineer

Oct. 2021 – May 2024

*ZDN Team* (Zalo Content Delivery Network): caching service serve millions request

- Implement scheduled `ScheduledThreadPool` for monitoring stats execution.
- Built admin dashboard with server visualization via `amchart`, `p5.js` in NextJS.
- Upgrade internal Golang package for Apache Thrift

**OpenHuman** ([openhuman.ai](https://openhuman.ai)), Creator

May 2024 – Present

- Join GENE Research Team ([GENEA Leaderboard](#), [HEMVIP](#)): Designed and built a full-stack, production-ready leaderboard system to evaluate multiple AI gesture generation models, integrating user study workflows via Prolific.

## </> Projects

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**MillionScope** (End-to-end ownership: [Code](#))

[millionscope.com](https://millionscope.com)

- Finetune GPT-OSS 20B via OpenAI-Cookbook with custom dataset.
- Experiment post-training RLHF chat response via [huggingface/trl](#), evaluate perplexity with [huggingface/evaluate](#)
- Develop full-stack both UI (Nextjs, Chat SDK, Cloudflare Pages) and backend LLM serving with SGLang on Thunder Compute, Cloudflare AI

**ParrotSpeech** (End-to-end ownership: [Code](#))

[parrotspeech.pages.dev](https://parrotspeech.pages.dev)

- Crawl speech dataset with `yt-dlp` youtube channels. Using `Chunkformer` to transcribe, preprocessing dataset.
- Fine-tune F5-TTS with Vietnamese speech and quantization QLoRA with `bitsandbytes`
- Export weights and serving via `onnx-runtime` and `transformers.js`

**DeepGesture** ([Code](#)): Convert euler rotation motion to quaternion-space, develop Diffusion Transformer (DiT) for generate MFCC coefficient (Similar MFCC speech but for skeleton motion movement).

[deepgesture.pages.dev](https://deepgesture.pages.dev)

**SurveyWithCode** ([Code](#)): PapersWithCode alternative using D3.js visualize taxonomy survey, provide DeepLearning knowledge.

[surveywithcode.pages.dev](https://surveywithcode.pages.dev)

**Others pet projects:** [Realistic Digital Human Render Engine](#), [Game Pucca Runner](#): dijkstra algorithm path finding game; [3D Threejs Skeleton](#), [MetaPet](#) ([metapet.vercel.app/Code](#)), [cortexpod.com](#); [pithagon.com](#)

## 🌟 Certifications

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- [Critical Thinking Certification 2014](#), [Emotion Recognition Certification 2019](#), [Summer ML](#)
- Coursera: [Design Pattern](#), [TensorFlow Developer Professional](#) 2021
- **AWS:** [Solutions Architect - Associate](#), [Machine Learning - Specialty](#); [Solution Architect](#) 2021

## 💻 Technologies

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**Programming:** Python, Javascript, C++, CUDA Kernel, SQL, Bash

**Frameworks & Libraries:** PyTorch, SGLang, `transformers.js`, PEFT, TRL, `bitsandbytes`, [confident-ai/DeepEval](#)

**Fine-Tuning:** QLoRA, SFT, RLHF, Knowledge Distillation

**Evaluation:** [perplexity](#), BLEU, ROUGE, Custom Metric Design (Toxicity, Relevance, Brand Voice)

**MLOps & Deployment:** Docker, Kubernetes, AWS SageMaker, `wandb`, MLflow, Grafana